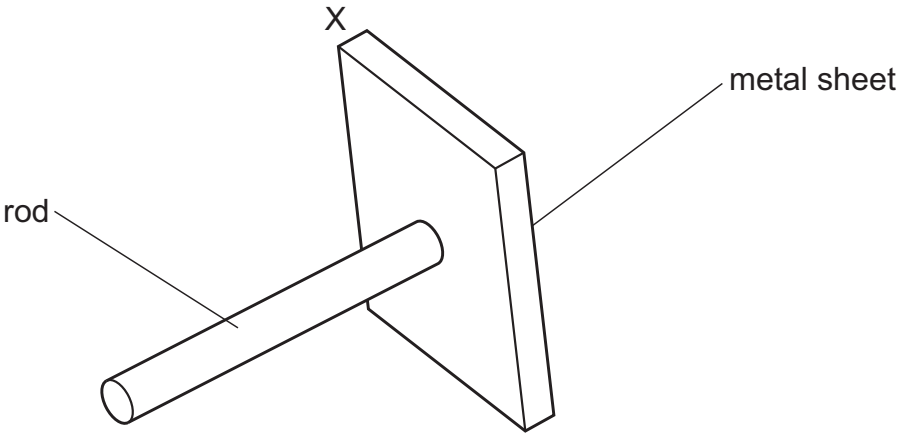


**2** Fig. 2.1 shows a square metal sheet of non-uniform density, with a thin wooden rod fixed at its centre. One of the corners of the sheet is labelled X.



**Fig. 2.1**

The rod has negligible mass. The mass of the metal sheet is 2.8 kg.  
The rod is supported so that the rod is horizontal and the metal sheet is vertical.

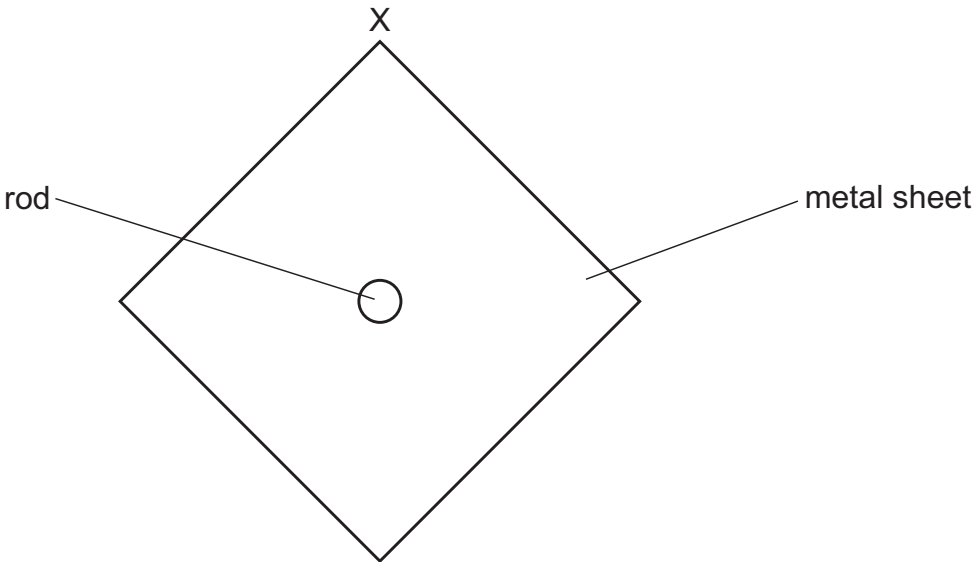
**(a)** Define the torque of a couple.

.....

.....

..... [2]

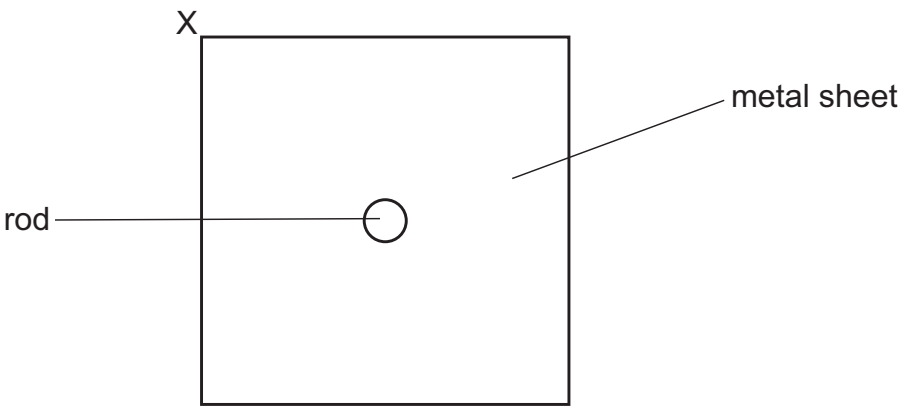
**(b)** When the rod is supported in such a way that it can rotate freely within its support, the sheet hangs in equilibrium with point X vertically above the rod, as shown in Fig. 2.2.



**Fig. 2.2**

On Fig. 2.2, draw a line to indicate the range of possible positions for the centre of gravity of the metal sheet. [1]

**(c)** When a torque of 3.3 N m is applied to the rod, the sheet is held in equilibrium with two of its edges horizontal, as shown in Fig. 2.3. Point X is at the top-left corner.



**Fig. 2.3**

**(i)** Explain whether the torque applied to the rod to hold the sheet in equilibrium is clockwise or anticlockwise.

.....

..... [1]

**(ii)** Show that the centre of gravity of the sheet has a horizontal displacement of 0.12 m from the rod.

[1]

**(d)** The square metal sheet has an average density of 3000 kg m<sup>-3</sup> and a uniform thickness of 4.0 mm.

Show that the side length of the sheet is 0.48 m.

[3]

**(e)** Use the answer in **(b)** and the information in **(c)** and **(d)** to determine the position of the centre of gravity of the sheet. Indicate this position on Fig. 2.3 with a point labelled Y. [2]